

B.S 2nd YEAR PHYSICS

LIST OF EXPERIMENTS

1. DETERMINE A SMALL UNKNOWN RESISTANCE BY CARY FOSTER BRIDGE.
2. DETERMINE THE UNKNOWN HIGH RESISTANCE USING AN RC CIRCUIT AND A NEON LAMP.
3. CONVERSION OF GALVANOMETER INTO AMMETER.
4. CONVERSION OF GALVANOMETER INTO VOLTMETER.
5. STUDY A REJECTOR CIRCUIT AND FIND THE VALUE OF INDUCTANCE.
6. STUDY AN ACCEPTER CIRCUIT AND FIND THE VALUE OF INDUCTANCE.
7. STUDY THE CHARACTERSTICS OF A GIVEN TRANSISTOR IN COMMON EMITTER MODE.

EXPERIMENT No: 01

OBJECT: DETERMINE A SMALL UNKNOWN RESISTANCE BY CAREY FOSTER BRIDGE.

APPARATUS:

DC power supply, galvanometer, resistance boxes, copper strip.

FORMULA:

$$R = S + (\ell_2 - \ell_1)\rho$$

Here

S = is the fractional resistance.

ℓ_1 = is the length of wire from end X to the null point when S is in the gap GH.

ℓ_2 = is the length of wire from end X to the null point when S is in the gap AB.

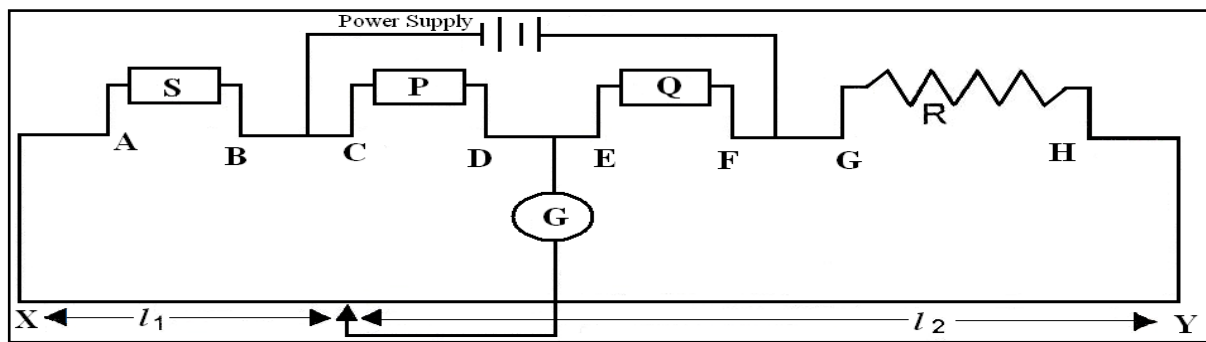
ρ = is the resistance per cm of the meter bridge wire given as:

$$\rho = \frac{S}{|(\ell_2 - \ell_1)|}$$

Here

ℓ_1 = is the length of wire from end X to the null point when S is in the gap AB.

ℓ_2 = is the length of wire from end X to the null point when S is in the gap GH.



PROCEDURE:

Determination of Resistance per cm of Wire

- 1) Make circuit connection as shown above, such that the resistance S is connected across the gap AB and the copper strip across the gap GH and two resistance boxes P & Q are connected across inner gaps CD and EF respectively.
- 2) Now take out keys from resistance boxes P and Q so that resistance of nearly equal values are introduced into the circuit and find out null point through galvanometer.
- 3) Measure the length ℓ_1 and ℓ_2 of the wire.
- 4) Repeat the above procedure for different set of readings.

Determination of Unknown Resistance

- 1) Keeping the resistance S constant, replace the copper strip with the unknown resistance R such that R is connected across GH and S is connected across AB.
- 2) Now take out keys from resistance boxes P and Q into the circuit and find out null point through galvanometer.
- 3) Measure the length ℓ_1 and ℓ_2 of the wire.
- 4) Repeat the above procedure for different set of readings.

OBSERVATIONS:**Determination of Resistance per cm of Wire**

Obs No.	Resistance (Ω)		Fractional Resistance S (Ω)	Length of Wire from X to null point l_1 (cm)	Length of Wire from null point to Y $l_2 = (100 - l_1)$ (cm)
	P	Q			
1.					
2.					
3.					

Determination of Unknown Resistance

Obs No.	Resistance (Ω)		Fractional Resistance S (Ω)	Length of Wire from X to null point l_1 (cm)	Length of Wire from null point to Y $l_2 = (100 - l_1)$ (cm)
	P	Q			
1.					
2.					
3.					

CALCULATIONS:

Calculate the values of resistance per cm of wire and small resistance from given formula.

RESULT:

The value of resistance per cm of wire is found to be _____ Ohms /cm.

The value of unknown resistance is found to be _____ Ohms.

REFERENCES:

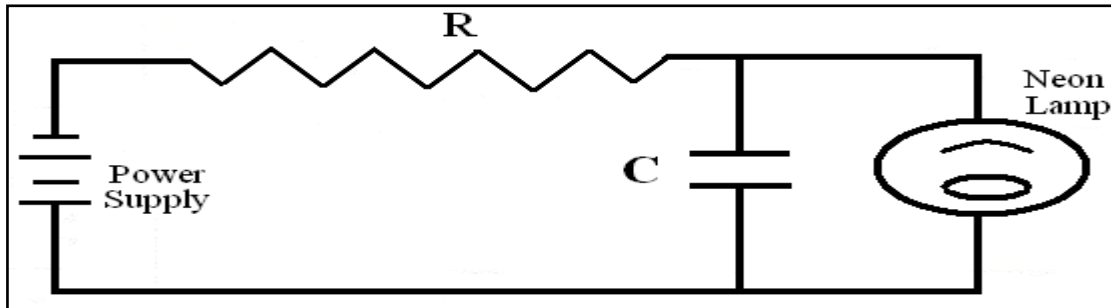
- 1) Practical Physics by Worsnop & Flint.
- 2) Physics by Halliday, Resnick and Krane.
- 3) Basic Electronics by Bernard Grob.
- 4) Electronic Devices by Floyd.
- 5) Electronic Principles by Malvino.

EXPERIMENT No: 02

OBJECT: DETERMINE THE UNKNOWN HIGH RESISTANCE USING “RC” CIRCUIT AND A NEON LAMP.

APPARATUS:

DC voltage source, High value resistances, Capacitor, Neon Lamp, Stop Watch.



PROCEDURE:

1. Make the circuit connections as shown in circuit diagram.
2. Start with resistance of 1 M ohm and note the time for 10 flashes.
3. Repeat step 2 by increasing the resistance by 1 M ohm each time.
4. Repeat step 2 with the three given unknown resistances (X_1 , X_2 , X_3).
5. Plot a graph between time for 1 flash and known resistance, which should be a straight line.
6. From the graph, read the value of unknown resistance with the help of its T value.

OBSERVATIONS:

Obs No	Resistance R (M Ω)	Time for 10 Flashes T (sec)				Time for 1 flash T/10 (sec)
		1	2	3	Mean	
1.						
2.						
...						
8.						

For determination of Unknown Resistance

Obs No	Time for 10 Flashes T (sec)				Time for 1 flash T/10 (sec)	Unknown Resistance (M Ω)
	1	2	3	Mean		
1.						X_1
2.						X_2
3.						X_3

GRAPH:

Plot a graph between known resistances (R) and time for 1 flash and obtain the unknown resistance values.

RESULT:

The unknown high resistances are found to be _____, _____, _____ M Ω .

REFERENCES:

- 1) Basic Electronics by Bernard Grob.
- 2) Physics by Halliday, Resnick and Krane.

EXPERIMENT No: 03

OBJECT: CONVERSION OF GALVANOMETER INTO AN AMMETER.

APPARATUS:

Power supply (or battery), Rheostat, Galvanometer, Shunt resistance, Ammeter.

FORMULA:

$$R_s = \frac{(R_M \times I_M)}{(I_T - I_M)}$$

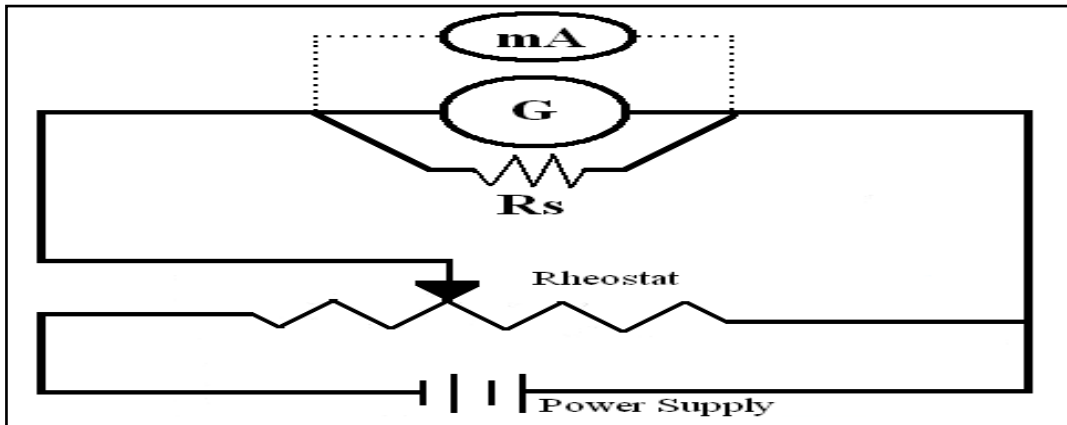
Here,

R_s = is value of shunt resistance in ohms.

R_M = is internal resistance of galvanometer = 120Ω

I_T = is applied current in amperes.

I_M = is full scale current rating of the meter moment in amperes = 0.3 mA



PROCEDURE:

- 1) Connect the circuit as shown in circuit diagram.
- 2) Connect the ammeter and adjust the rheostat such that ammeter reads 5mA, then for this constant current calculate the value of R_s .
- 3) Connect the galvanometer in circuit instead of ammeter and introduce calculated value of resistance in parallel with the galvanometer. In this condition Galvanometer show full scale deflection it means it reads 5mA of current.
- 4) Change on the constant current in step of 5mA and each time repeat step 2 & 3.

OBSERVATIONS:

Obs No	Applied Current I_T (mA)	Calculated value of Shunt Resistance (R_s) Ω	Observed value of shunt resistance Ω	Percentage Error
1.	5			
2.	10			
3.	15			
4.	20			
5.	25			

CALCULATIONS:

Calculate the value of shunt resistances and percentages of error for all applied currents.

RESULT:

Galvanometer was converted into Ammeter for current _____ mA

With shunt resistance calculated _____ Ω

Observed _____ Ω

Percentage error in result = $\frac{(\text{Calculated value} - \text{observed value})}{\text{Calculated value}} \times 100$

REFERENCES:

- 1) Basic Electronics by Grob.
- 2) Practical Physics by Worsnop & Flint.
- 3) Magnetism and Electricity by D. M. Vasudeva.

EXPERIMENT No: 04

OBJECT: CONVERSION OF GALVANOMETER INTO VOLTMETER.

APPARATUS:

Power supply (or battery), Rheostat, Galvanometer, High resistance box, Voltmeter.

FORMULA:

$$R_S = \left[\frac{V_T}{I_M} - R_M \right]$$

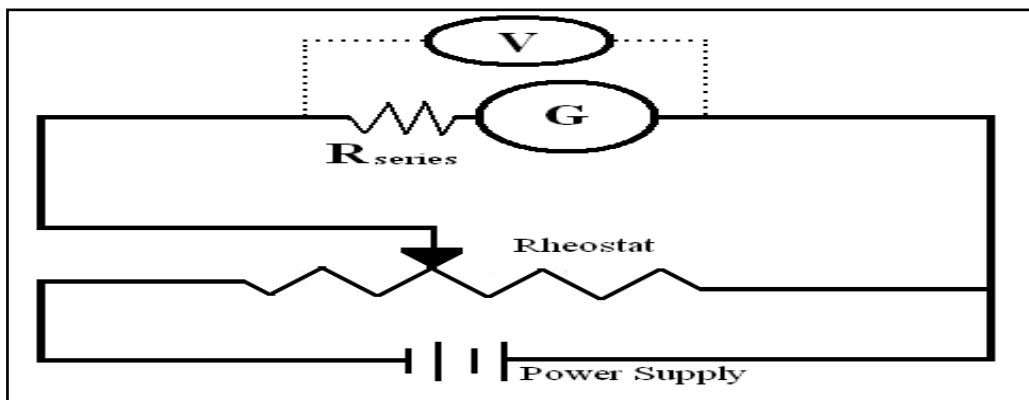
Here,

R_A = is value of resistance in ohms.

R_M = is internal resistance of galvanometer = 120Ω

V_T = is applied voltage in volts.

I_M = is full scale current rating of the meter moment in amperes = 0.3 mA



PROCEDURE:

- 1) Connect the circuit as shown in circuit diagram.
- 2) Connect the voltmeter and adjust the rheostat such that voltmeter reads 1V, then for this constant voltage calculate the value of R_A .
- 3) Connect the galvanometer in circuit instead of voltmeter and introduce calculated value of resistance in series with the galvanometer. In this condition Galvanometer show full scale deflection it means it reads 1V of voltage.
- 4) Change on the constant voltage in step of 1V and each time repeat step 2 & 3.

OBSERVATIONS:

Obs No	Applied Voltage V_T (V)	Calculated value of Series Resistance (R_A) Ω	Observed value of series resistance Ω	Percentage Error
1.	1			
2.	2			
3.	3			
4.	4			
5.	5			

CALCULATIONS:

Calculate the value of series resistances and percentages of error for all applied voltages.

RESULT:

Galvanometer was converted into Voltmeter for voltage _____ V.

With series resistance calculated _____ Ω .

Observed _____ Ω .

Percentage error in result = $\frac{(\text{Calculated value} - \text{observed value})}{\text{Calculated value}} \times 100$

REFERENCES:

- 1) Basic Electronics by Grob.
- 2) Practical Physics by Worsnop & Flint.
- 3) Magnetism and Electricity by D. M. Vasudeva.

EXPERIMENT No: 05

OBJECT: STUDY A REJECTOR CIRCUIT AND FIND THE VALUE OF INDUCTANCE.

APPARATUS:

Oscilloscope, Frequency generator, capacitor, resistor and inductor in a circuit box.

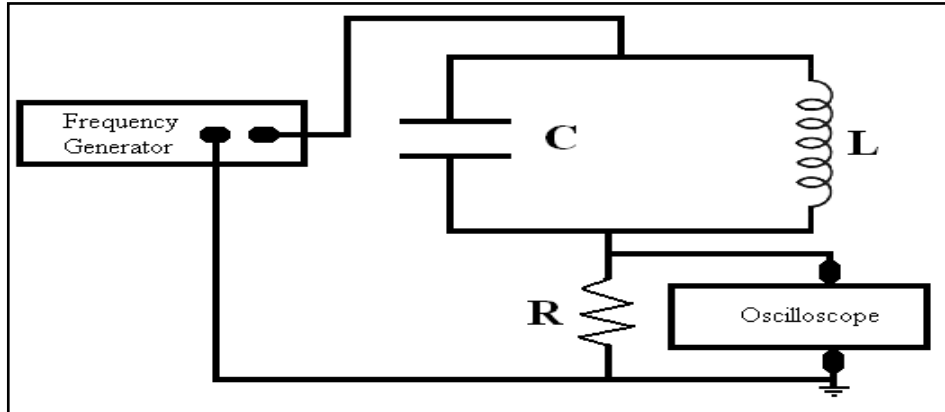
FORMULA:

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

C = is capacity of Capacitor = 15.7×10^{-9} F

f_r = is the resonant frequency.

L = is the required Inductance.



PROCEDURE:

1. Check the connection of circuit then turn on the main power of oscilloscope and function generator.
2. By setting 10 KHz frequency on function generator note down the peak-to-peak output on oscilloscope.
3. Repeat the above step for given frequencies.
4. Plot a graph between Log F and peak-to-peak output of oscilloscope.
5. From graph find out resonant frequency (frequency of minimum voltage), taking antilog.
6. Calculate the value of inductance (L) from given formula.

OBSERVATIONS:

Obs No	Frequency f (Hz)	Peak-Peak output voltage on oscilloscope (V)	Log f
1.			
2.			
.			
.			
12.			

GRAPH:

Plot a graph of peak-peak output voltage verses log f.

RESULT:

Value of resonant frequency (f_r) from graph = _____ Hertz.

Value of inductance (L) from the formula = _____ Henry= _____ mH.

REFERENCES:

1. Basic Electronics by Bernard Grob.
2. Physics by Halliday, Resnick and Krane.
3. Electronic Devices by Floyd.
4. Electronic Principles by Malvino.

EXPERIMENT No: 6

OBJECT: STUDY AN ACCEPTOR CIRCUIT AND FIND THE VALUE OF INDUCTANCE.

APPARATUS:

Oscilloscope, Frequency generator, capacitor, resistor and inductor in a circuit box.

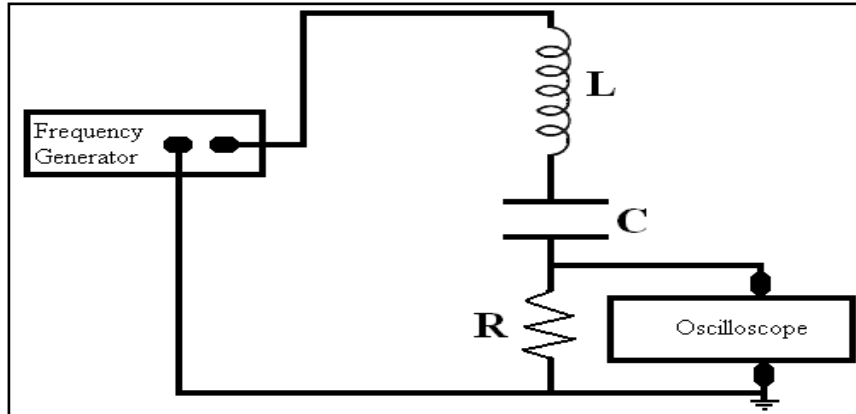
FORMULA:

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

C = is capacity of Capacitor = 0.01×10^{-6} F

f_r = is the resonant frequency.

L = is the required Inductance.



PROCEDURE:

1. Check the connection of circuit then turn on the main power of oscilloscope and function generator.
2. By setting 100 Hz frequency on function generator note down the peak-to-peak output on oscilloscope.
3. Repeat the above step for given frequencies.
4. Plot a graph between Log F and peak-to-peak output of oscilloscope.
5. From graph find out resonant frequency (frequency of maximum voltage), taking antilog.
6. Calculate the value of inductance (L) from given formula.

OBSERVATIONS:

Obs No	Frequency f (Hz)	Peak-Peak output voltage on oscilloscope (V)	Log f
1.			
2.			
.			
.			
25.			

GRAPH:

Plot a graph of peak-peak output voltage verses log f.

RESULT:

Value of resonant frequency (f_r) from graph = _____ Hertz.

Value of inductance (L) from the formula = _____ Henry.

REFERENCES:

1. Basic Electronics by Bernard Grob.
2. Physics by Halliday, Resnick and Krane.
3. Electronic Devices by Floyd.
4. Electronic Principles by Malvino.

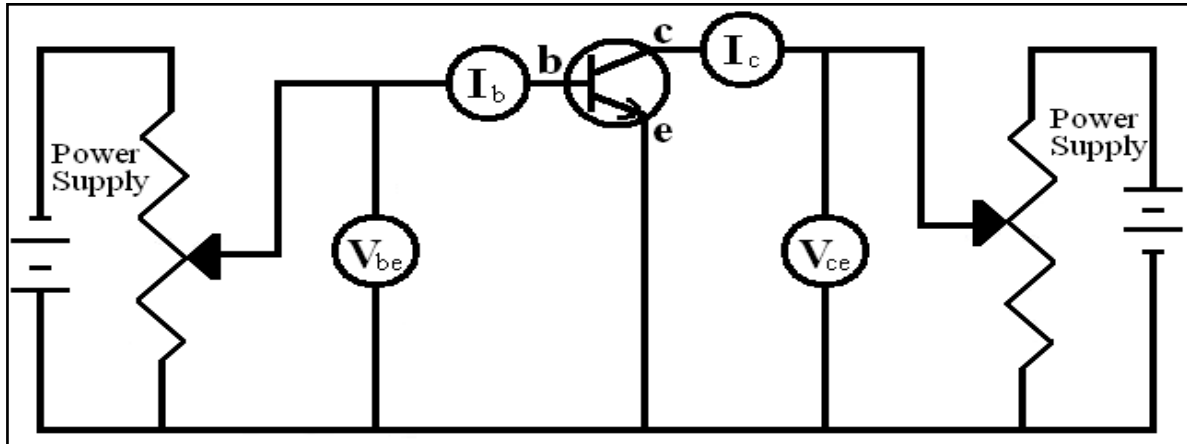
EXPERIMENT No: 07

OBJECT: STUDY THE STATIC CHARACTERISTICS OF A GIVEN TRANSISTOR IN COMMON EMITTER MODE.

PRECAUTIONS: Do **NOT** exceed by a maximum of 12V for collector Voltage. Also do not exceed the base circuit supply voltage by 3V.

APPARATUS:

Small power supply, Voltmeter (0-3V) and ammeter for base circuit, two way keys, transistor AD161 mounted with the protective base resistance.



PROCEDURE:

- 1) Connect the above circuit carefully observing the polarities of the instruments and setting the rheostat and potentiometer for minimum output.
- 2) Use the two way key to first note the voltage to be applied by filling gap 1, and apply this voltage to the circuit by filling gap 2. Gap 1 and Gap 2 both must not be closed at the same time because the low impedance voltmeter loads the circuit.

Input Characteristics

- 3) Study of I_b versus V_{be} for different values of V_{ce} carefully observing the instructions about the two ways key given above.
- 4) Apply a V_{ce} of 0V, then choose $V_{be} = 0V$, and apply it to base, observing the base current keeping V_{ce} constant at 0V and increasing V_{be} to 1.6V in steps of 0.2 V each time noting I_b .
- 5) Now take two more sets of observation as described above, but by keeping V_{ce} at 2V and 4V respectively.

Output Characteristics

- 6) Study of I_c versus V_{ce} for different values of I_b . again carefully observing the instructions.
- 7) Set $I_b = 20$ microampere. This can be achieved opening the gap 2 of key K_1 . Note the collector current by gradually increasing the value of V_{ce} in steps of 1V up to a max. of 7 V.
- 8) Take two more sets of observation by keeping I_b at approximately 40 and 60 microamperes.

OBSERVATIONS:

INPUT CHARACTERISTICS

Obs No	$V_{ce} = 0V$		$V_{ce} = 2V$		$V_{ce} = 4V$	
	V_{be}	I_b	V_{be}	I_b	V_{be}	I_b
1.						
2.						
3.						
4.						
5.						
6.						
7.						
8.						
9.						

OUTPUT CHARACTERISTICS

Obs No	$I_b = 20 \mu A$		$I_b = 40 \mu A$		$I_b = 60 \mu A$	
	V_{ce}	I_c	V_{ce}	I_c	V_{ce}	I_c
1.						
2.						
3.						
4.						
5.						
6.						

CALCULATIONS:

Draw the graphs for:

- Input characteristics between V_{BE} and I_B .
- Output characteristics between V_C and I_C .

RESULT:

Discuss the behavior of graphs.

REFERENCES:

- 1) Physics by Halliday, Resnick and Krane.
- 2) Electronic Devices by Floyd.
- 3) Electronic Principles by Malvino.